



Benjamin Hsieh

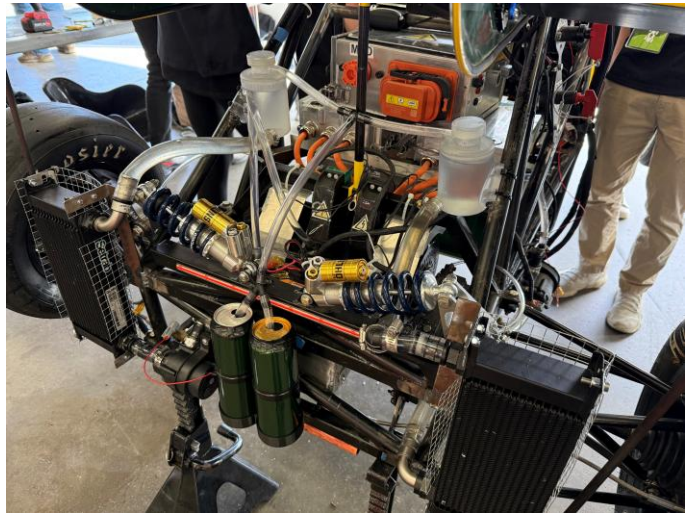
Engineering Portfolio

Dartmouth College: Mechanical Engineering

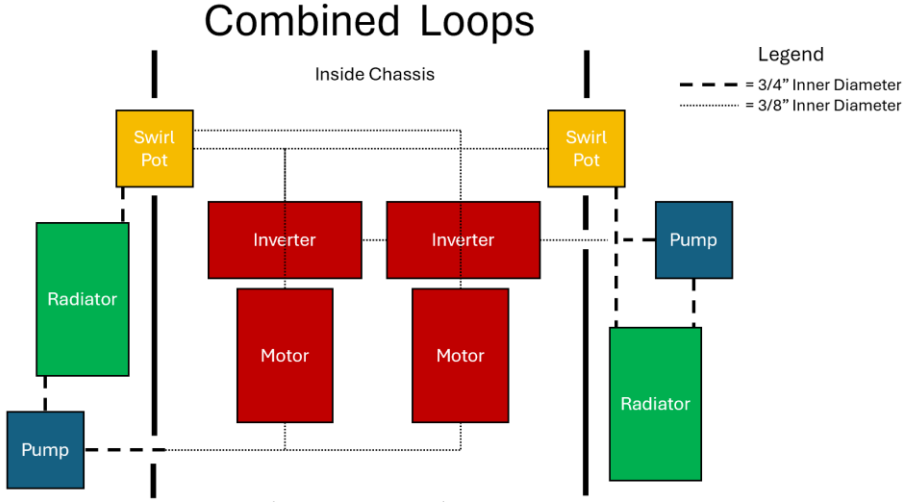
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Dartmouth Formula Racing: Cooling System

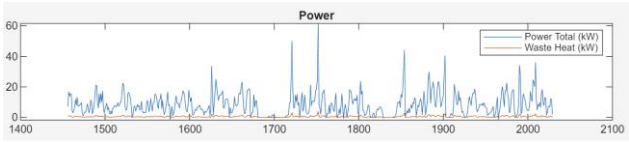
- Designed swirl pot component in SOLIDWORKS, eliminating trapped air and decreasing cooling system bleed time from 15+ minutes to 2
- Developed and conducted cooling loop bench tests, validating coolant flow rates for sufficient cooling with previous components
- Used first principle heat transfer equations to derive mass flow target for optimum heat rejection
- Initiated team wiki knowledge transfer for cooling system fundamentals

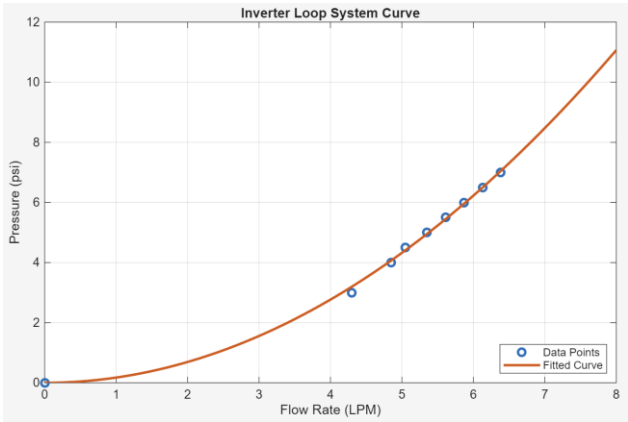


Final Cooling Loop Assembly

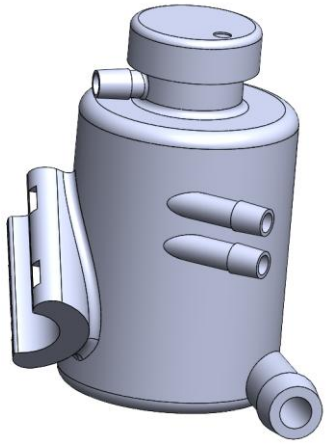


Cooling Loop Diagram





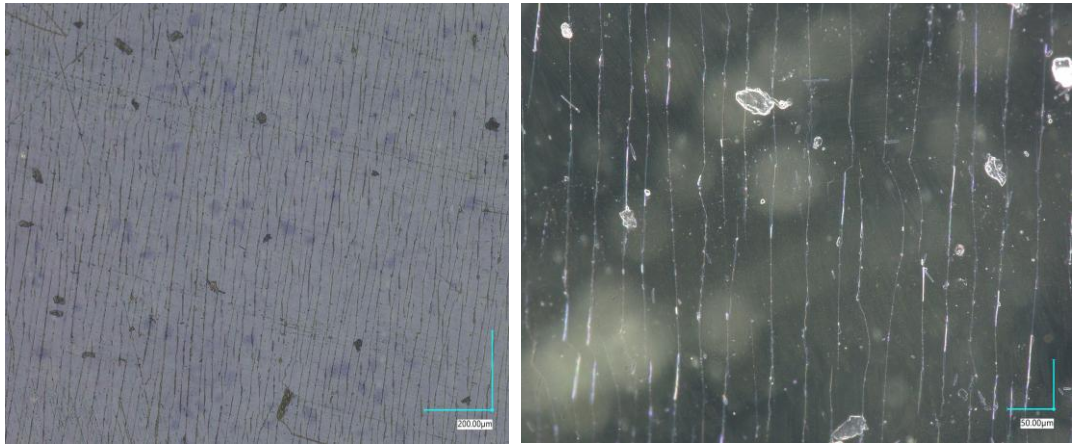
Waste Heat Visualization and System Curve Plotting



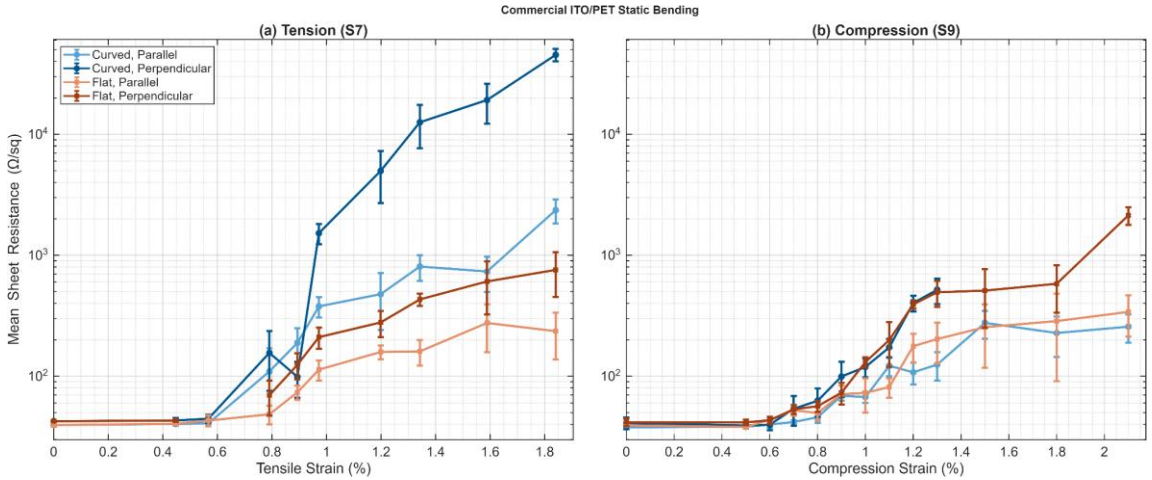
Swirl Pot Final Design

SENSE-Lab: Mechanical and Laser Treatment of Transparent Conductive Films

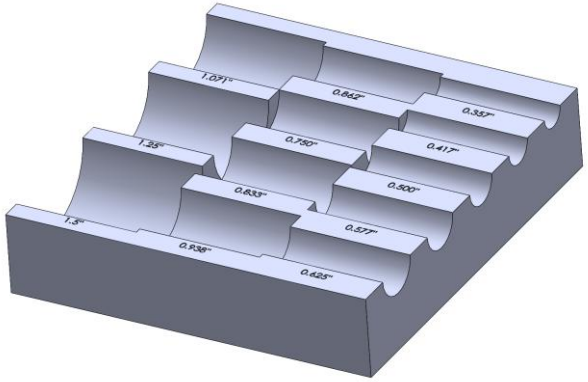
- Developed and executed tensile and compressive test procedures on thin conductive films to identify critical strain thresholds
- Created custom 3-D bending jig in SOLIDWORKS for compressive testing
- Investigated laser-induced crystallization of thin conductive films, decreasing electrical resistivity by 70%



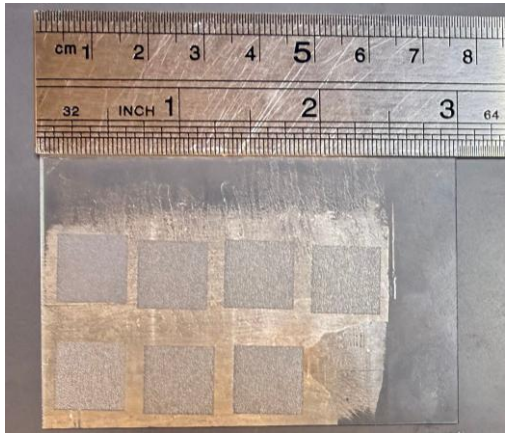
Optical Microscope Crack Visualization



Tensile and Compressive Strain v. Sheet Resistance



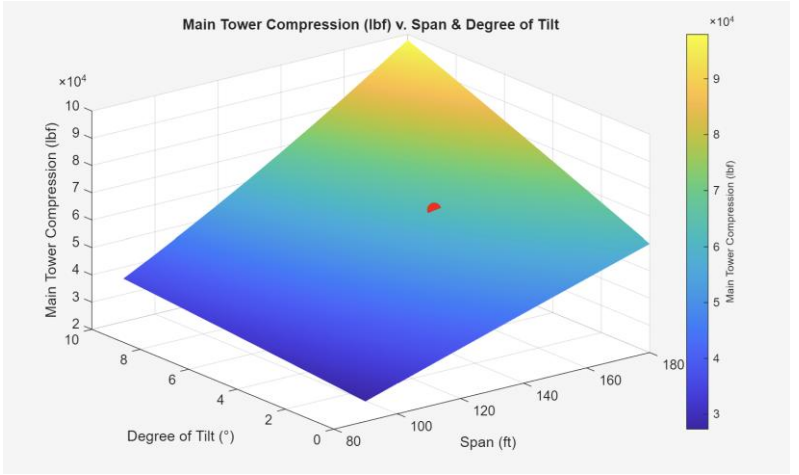
Compressive Bending Apparatus



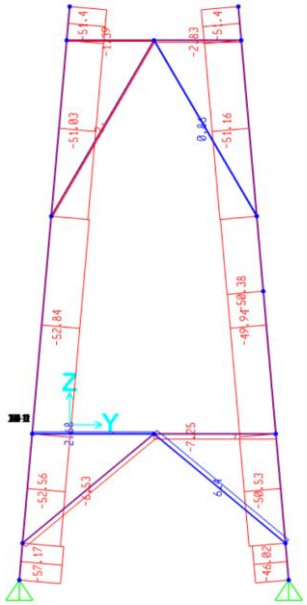
Initial Laser Annealment Trials

ENGS 71: Moosilauke Ravine Lodge Bridge

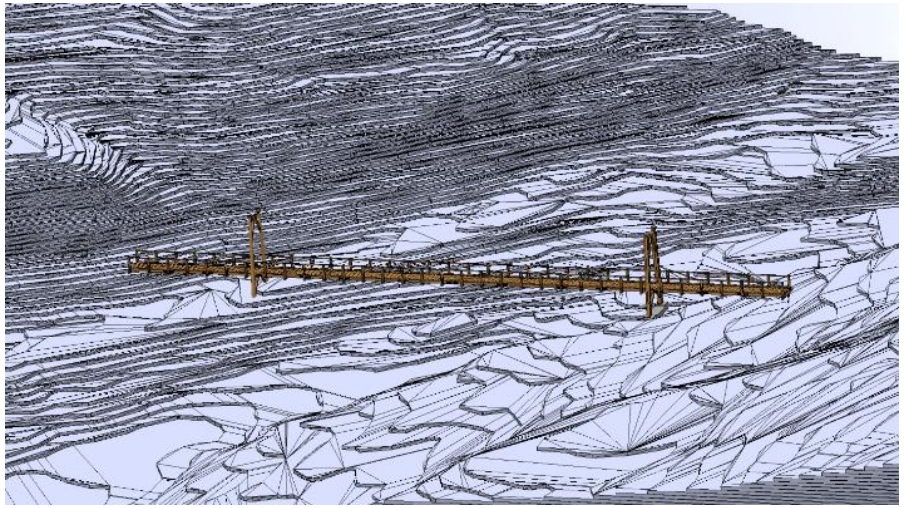
- Ran SAP2000 Analyses to determine axial, shear, and moment under loading conditions as specified by ASCE/SEI 7-22
- Consulted ACI, AWC, and HUD building codes, creating a custom steel bearing plate and concrete foundation model
- Delivered a comprehensive bill of materials for two tower designs



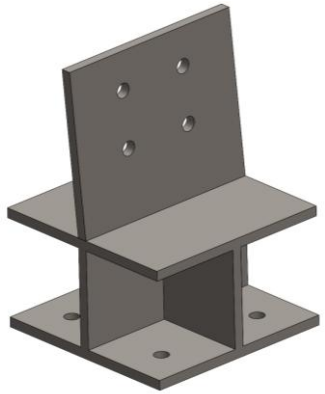
Preliminary Design
Compression Visualization



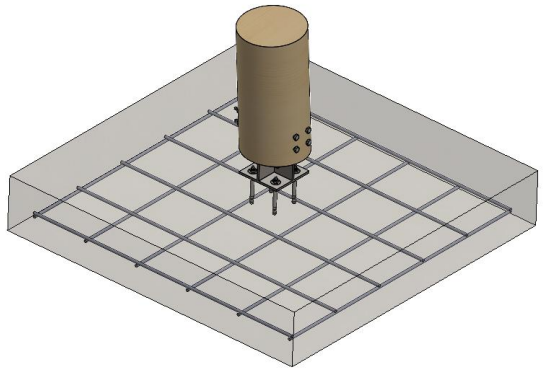
SAP2000 Analysis
Axial Forces



Final Bridge Design Render



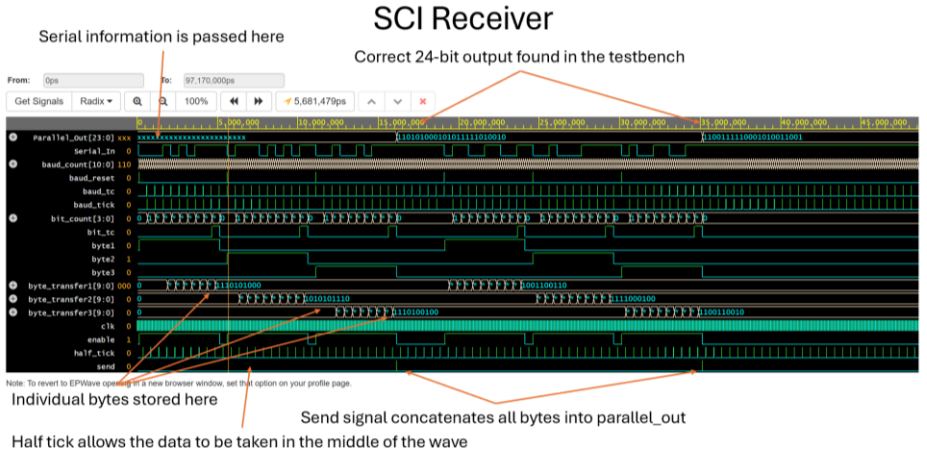
Custom Bearing Plate



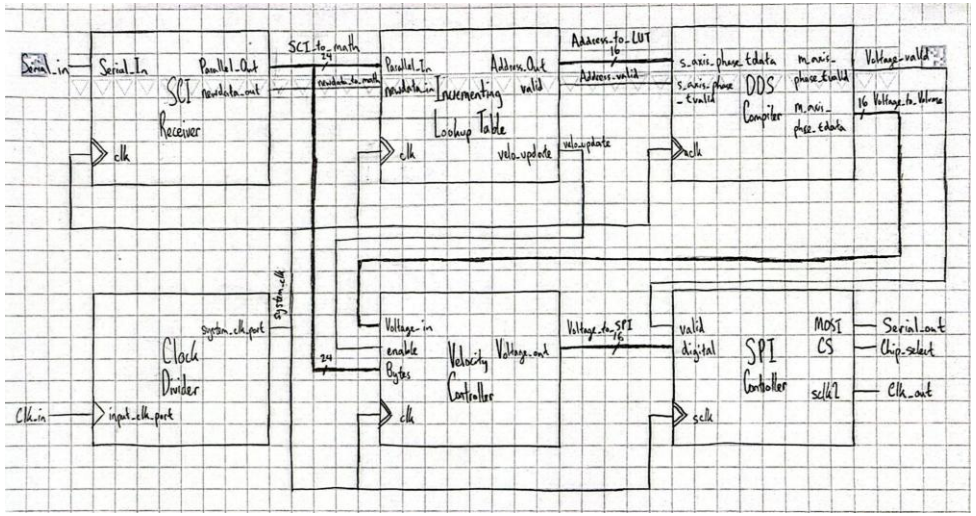
Rebar

ENGS 31: FPGA MIDI-to-Audio Synthesizer

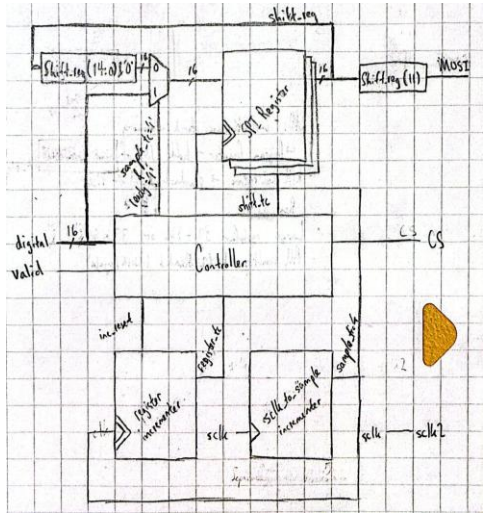
- Created an FPGA converting multi-note MIDI signals into velocity-sensitive audio output
- Executed top-down design and bottom-up implementation, creating six custom VHDL components spanning signal decoding, digital synthesis, and DAC output
- Validated components on the individual and system level in both digital simulation and hardware testing



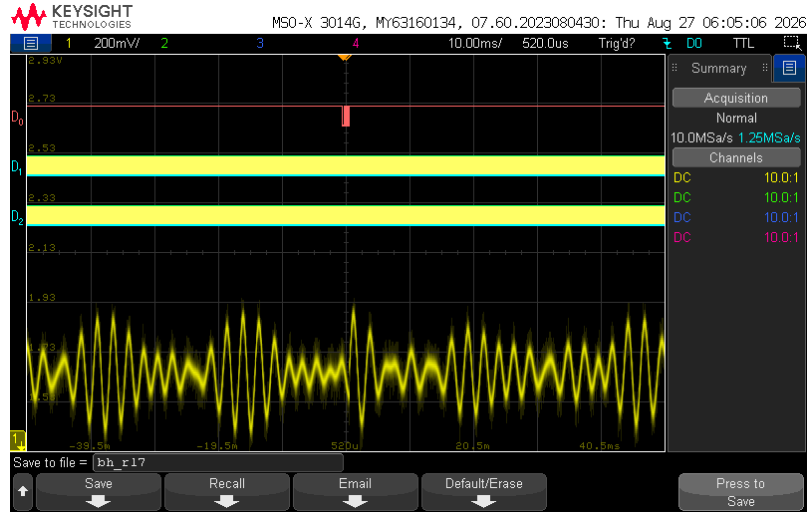
SCI Receiver Behavioral Simulation



Top Level Block Diagram



SPI Controller RTL Diagram



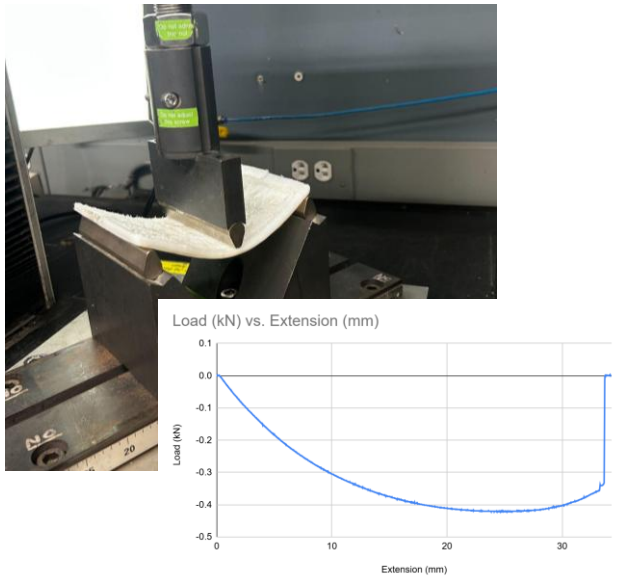
Hardware Implementation – Multi-note Input

ENGS 21: Recycled Plastic Molding Process

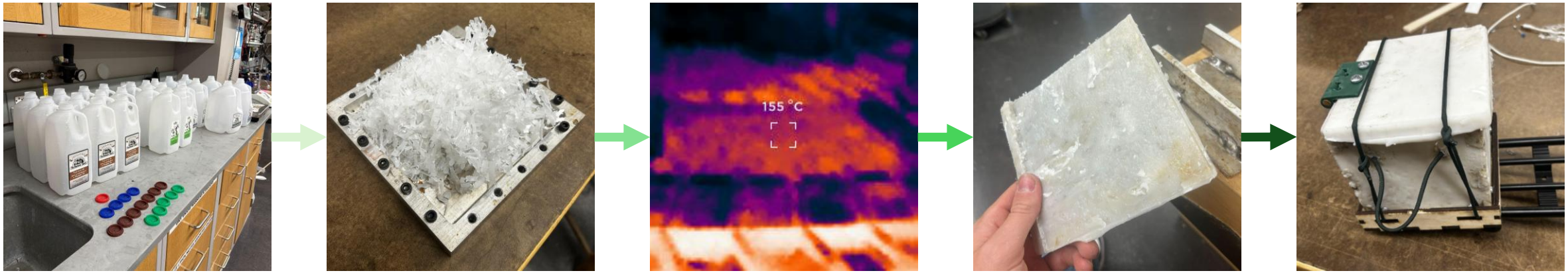
- Designed protocol for recycling, shredding, melting, and molding plastic for use in food transportation
- Conducted Instron structural material tests on melted plastics, ensuring an ability to withstand the proper loading conditions
- Collaborated with nonprofit Moving Windmills weekly via email and Google Meet to adhere to client specifications and procedural limitations



Modulus of Elasticity Testing



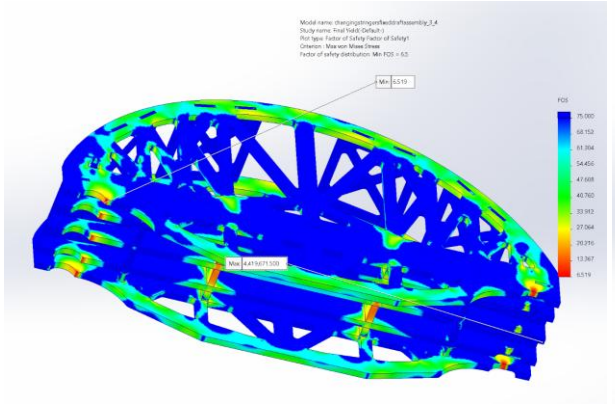
Flexural Modulus Testing



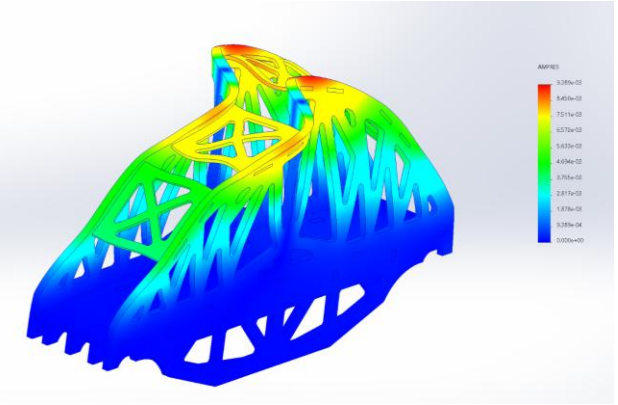
Plastic Molding Process

ENGS 33: Pedestrian Footbridge Modeling

- Designed 1:16 model pedestrian footbridge to withstand 1 kN load
- Completed SOLIDWORKS FEA for compressive loading and buckling, accurately determining the point of failure and method of failure
- Fabricated model, considering laser cutting tolerances, wood grain orientation, and gluing procedures



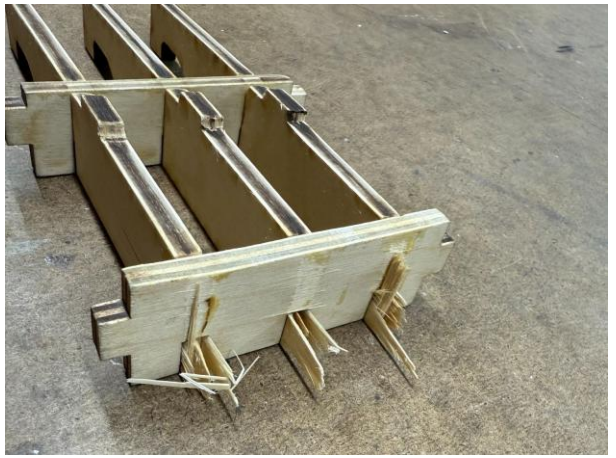
Compressive Loading FEA



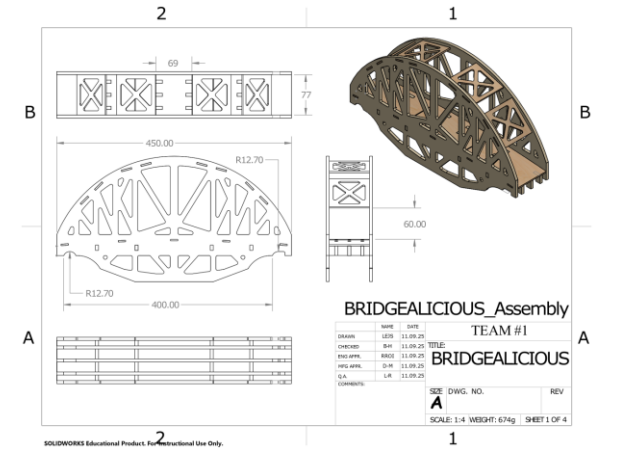
Buckling FEA



Final Assembly



Point of Failure Example



SOLIDWORKS Assembly